

ECSE 4560/ 6560

Modern Communication Systems

11

Lecture ~~10~~: OFDM CFO



Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: CFO
- Projects
 - Group formation [MCS Fall25 project.xlsx](#)
- HW3/Lab3 due Today Friday, 10/10
- HW4 will be posted soon.
- Exam 1: Friday Oct 24
- Exam 2: Friday Nov 21
- Final Project Presentation and Demos: Dec 9 and Dec 12
- Project Report and peer evaluation: Finals exam week

Open Book
No cheat lab

Extra Reading on Piazza

- Principles of Modern Wireless Communication Systems Theory and Practice; by Aditya K. Jagannatham, IIT Kanpur Book
 - Parts of Chapter 1,2,3,4,7
- An Adaptive Orthogonal Frequency Division Multiplexing Baseband Modem for Wideband Wireless Channels; by Jit Ken Tan, MIT Thesis
 - OFDM Decoding and Synchronization
 - Lab 4 Implementation

Today's Class: OFDM CFO

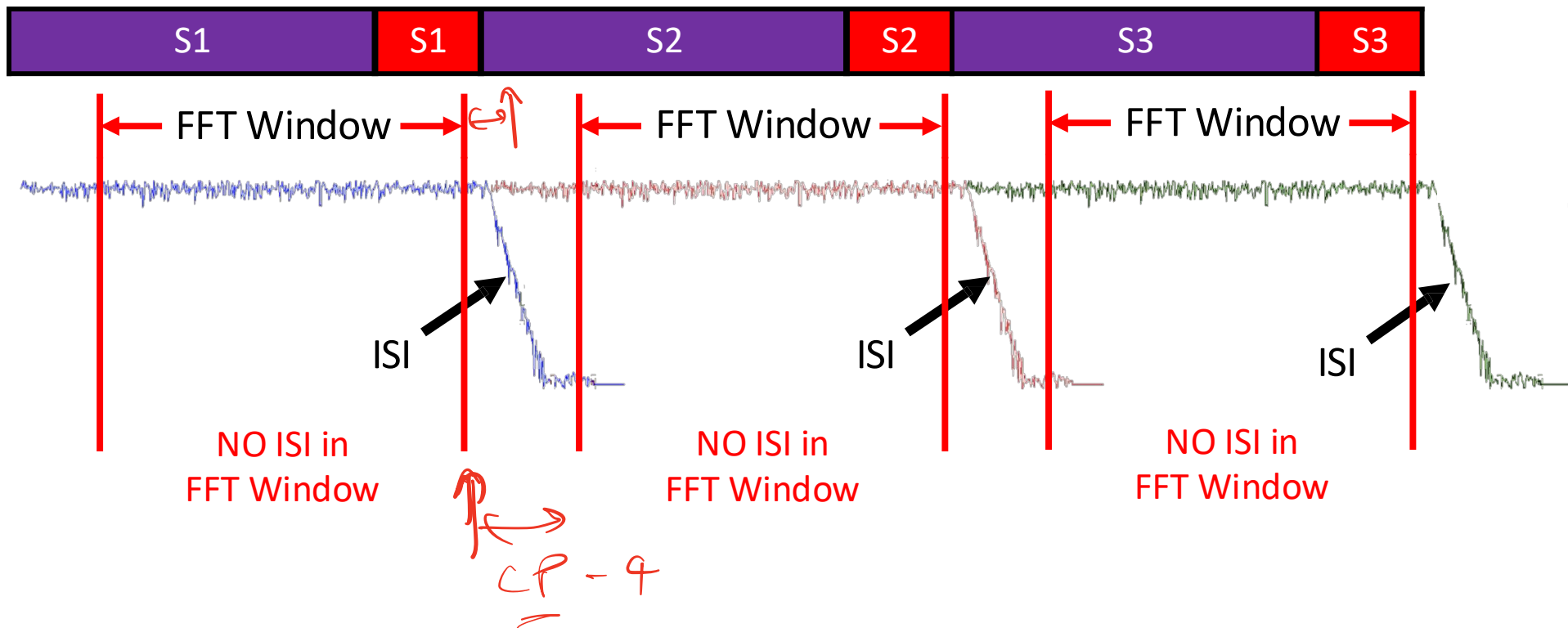
- OFDM Synchronization
 - Cyclic Prefix (revise)
 - Packet detection (revise)
 - CFO correction
 - Channel Estimation
 - SFO and Residual Phase correction

OFDM Cyclic Prefix

- ① Multipath
- ② Packet detection offset

Cyclic Prefix:

- Preserves orthogonality by allowing some misalignment in FFT Window
- Deals with Inter-Symbol-Interference



OFDM Cyclic Prefix

Cyclic Prefix:

 Preserves orthogonality by allowing some misalignment in FFT Window

 Deals with Inter-Symbol-Interference

 Overhead: Send $CP + N$ samples for every N samples

$$\text{Overhead} = \frac{CP}{CP + N}$$

e. g. WiFi 802.11n: $N = 64$, $CP = 16 \rightarrow \text{Overhead} = 20\%$

e. g. LTE: $N = 1024$, $CP = 72 \rightarrow \text{Overhead} = 6.5\%$

OFDM Cyclic Prefix



- Cyclic prefix is not a bullet proof solution.
- Can still end up misaligned!
- Need a way to ensure we detect the beginning of the packet correctly.
- If we do, CP will ensure that even if we are not accurate, we can still decode.

OFDM Packet Detection

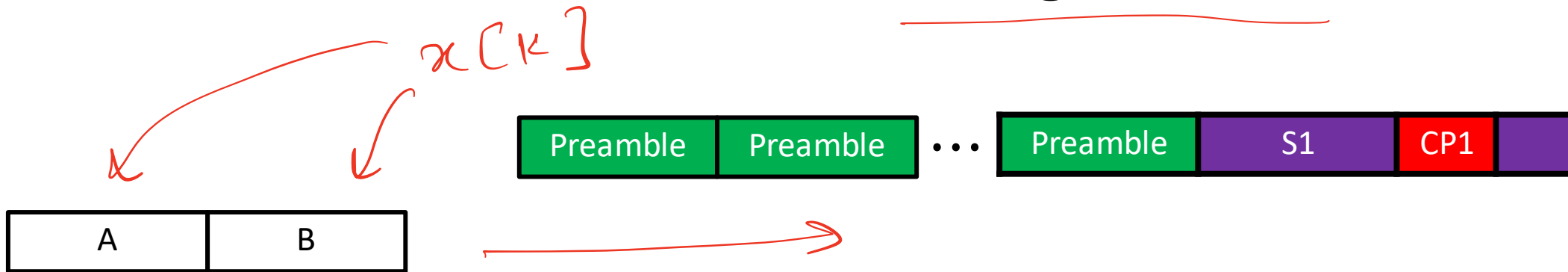
- Detect Beginning of packet to make sure we are within the CP
- Send Training Sequence: Preamble Symbols
- Preamble Symbols: Known Symbol Repeated at the beginning of packet



- Generally, 2 copies of preamble (LTS – Long Training Sequence) used in WiFi + 0.5 copy as cyclic prefix

Lo ~ 10 copies of STS short term sequence

OFDM Packet Detection: Sliding Window

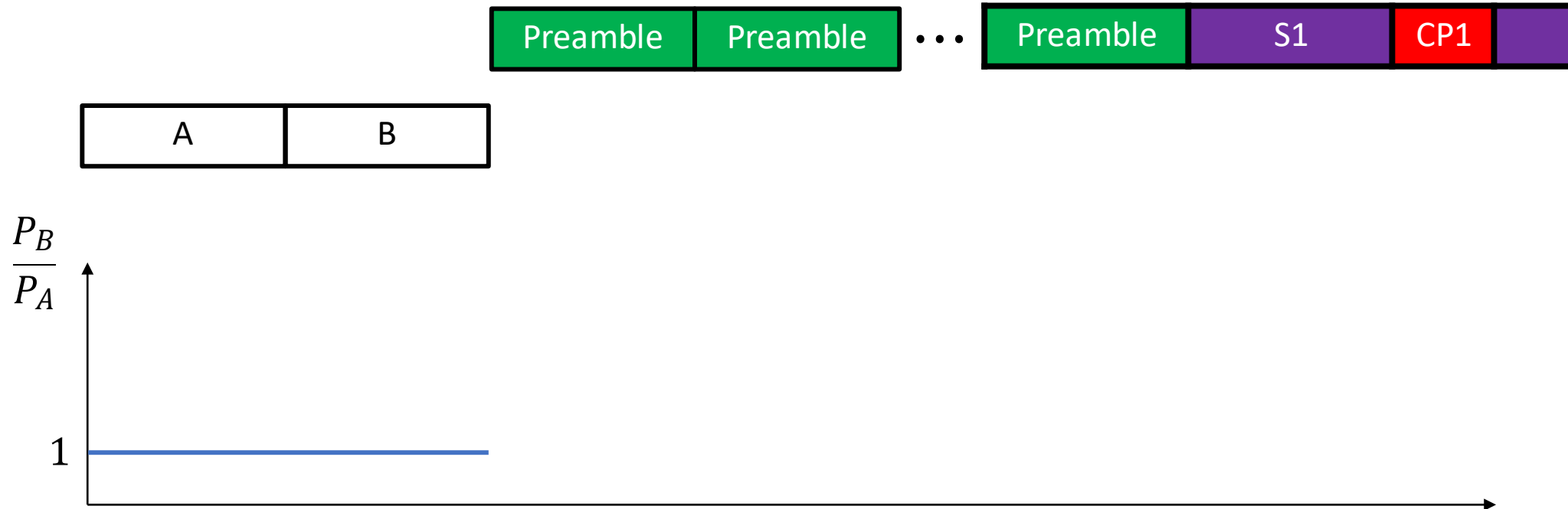


- Two windows of L ($2N$) samples each.

- Compute:

$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

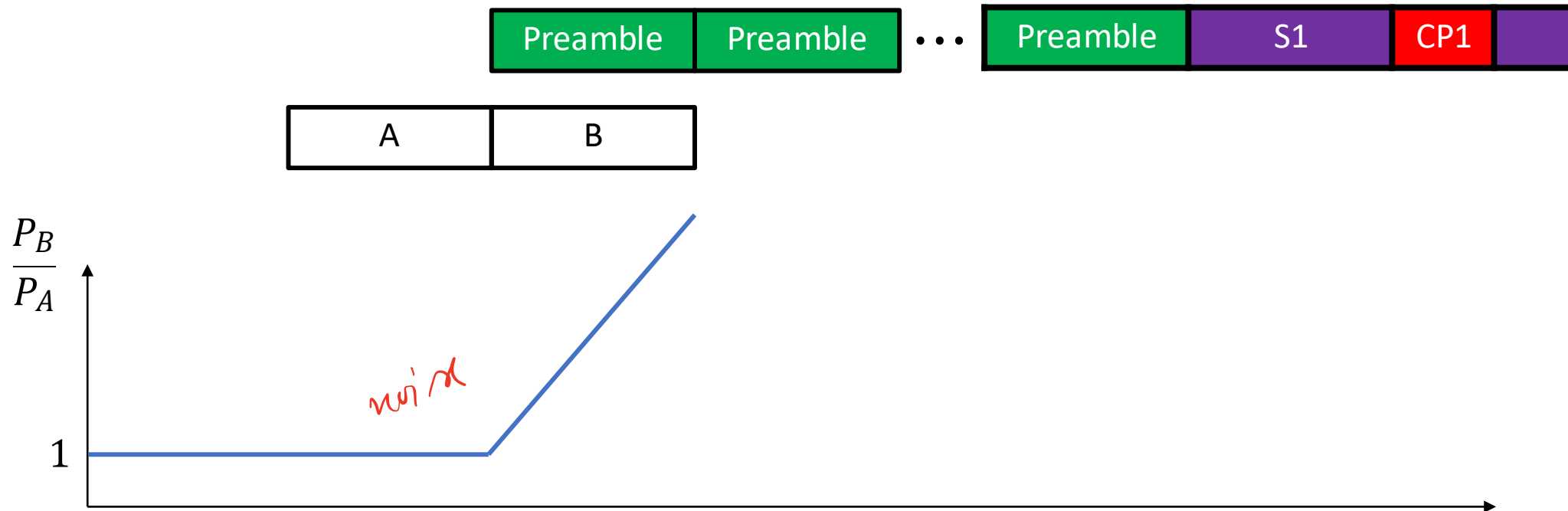
OFDM Packet Detection: Sliding Window



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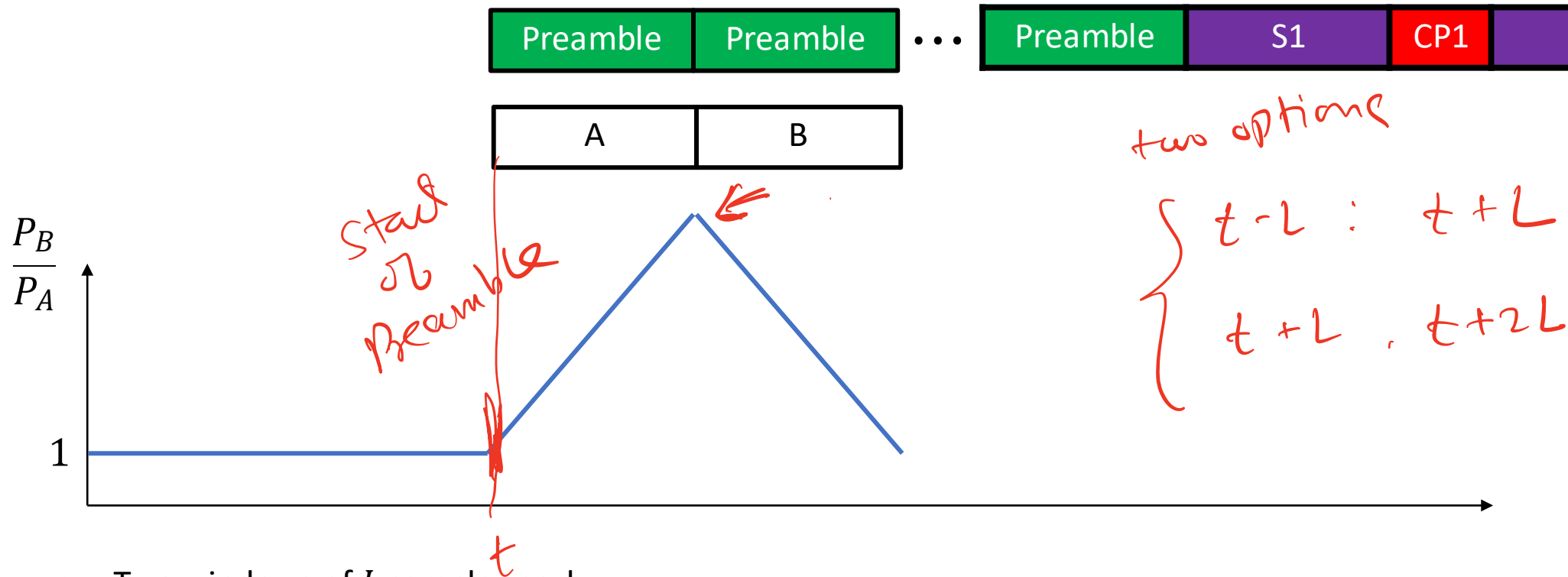
OFDM Packet Detection: Sliding Window



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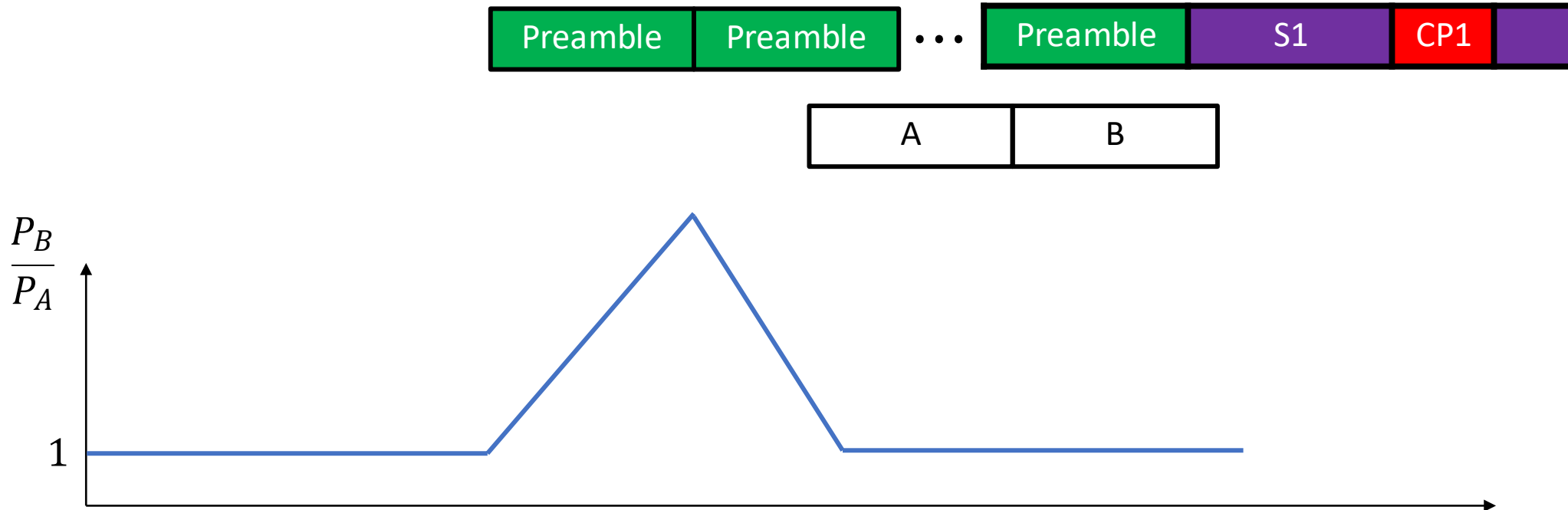
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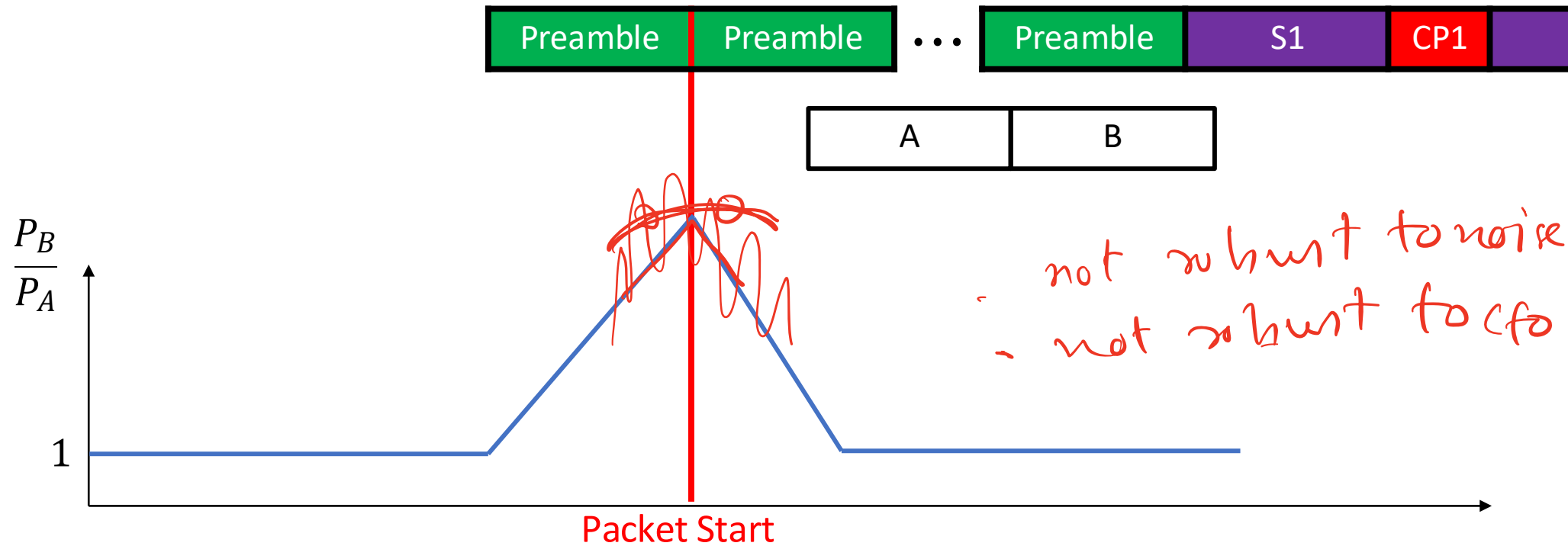
OFDM Packet Detection: Sliding Window



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OFDM Packet Detection: Sliding Window



- Two windows of L samples each.

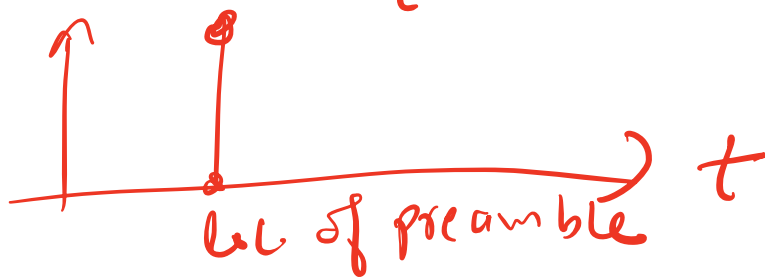
- Compute:
$$\frac{P_B}{P_A} = \frac{\sum_{k=t+L}^{t+2L} |x[k]|^2}{\sum_{k=t}^{t+L} |x[k]|^2}$$

OFDM Packet Detection: Cross Correlation

takes a PN sequence which is known to the receiver

$\Rightarrow$ Rx correlates the Rx data with known PN sequence

$\Rightarrow$ find peak when correlation is max $\Rightarrow$ PN seq aligns with transmitted preamble



Optimal OFDM Packet Detection: Schmid–Cox, 802.11 style (Reading Exercise - Piazza)

auto correlation.

at least
two OFDM
preamble

$$P(d) = \sum_{m=0}^{L-1} (r_{d+m}^* r_{d+m+L}) \quad (5)$$

↑ shifted by L

$$R(d) = \sum_{m=0}^{L-1} |r_{d+m+L}|^2$$

$$M(d) = \frac{|P(d)|^2}{(R(d))^2}. \quad (8)$$

How often we send preamble over time?

Ans: Channel coherence time

Doppler
freq
↓

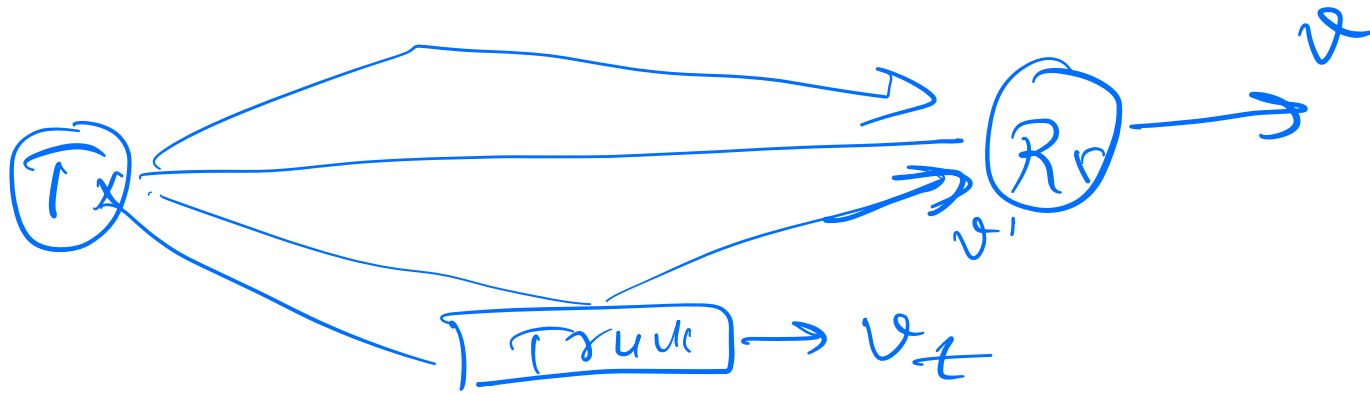
$$x(t) \longrightarrow y(t) \quad j 2\pi f_d t$$

$$= h(t) * x(t) e$$

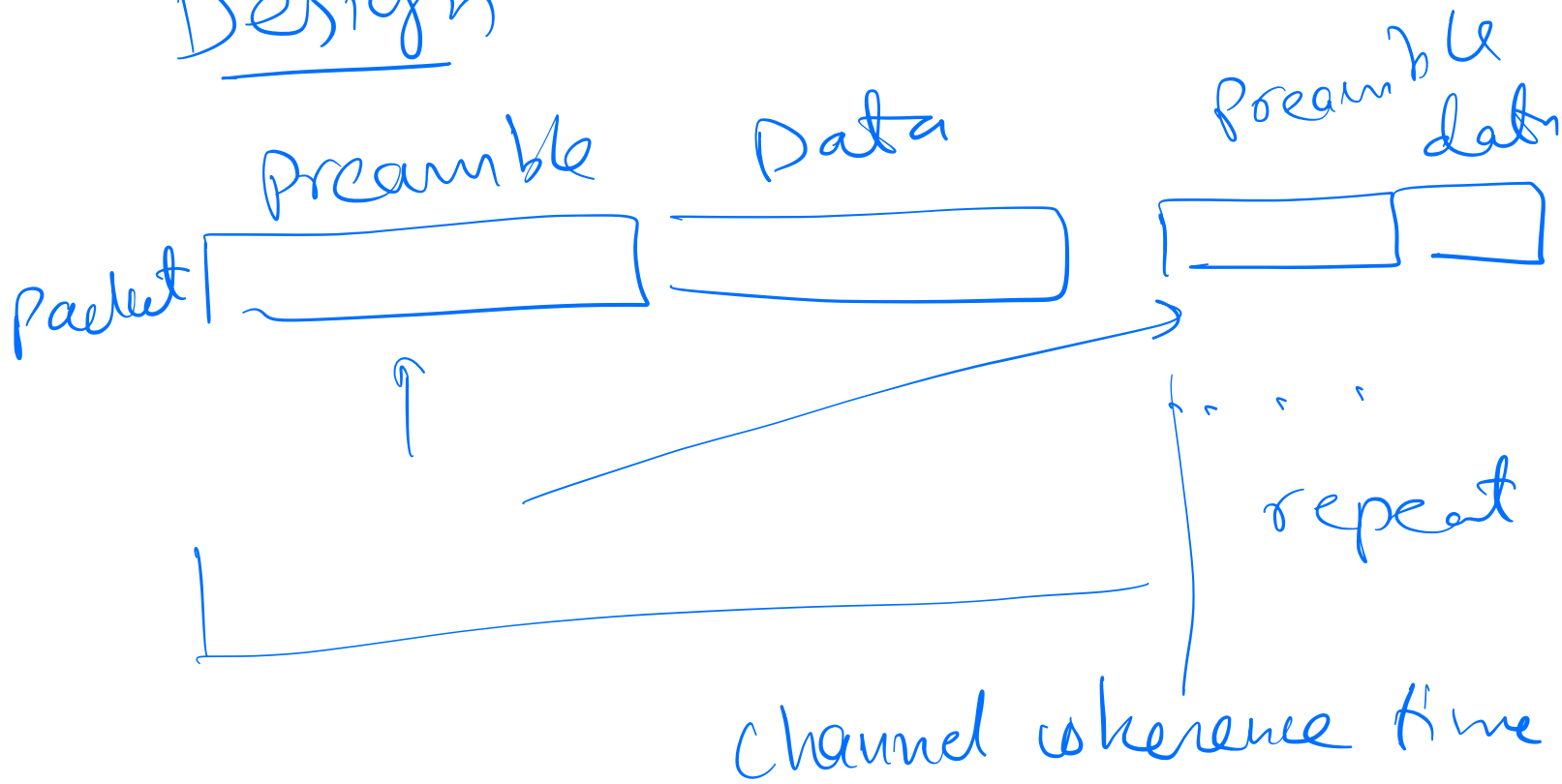
Doppler spread

- Diff path may have
diff Doppler

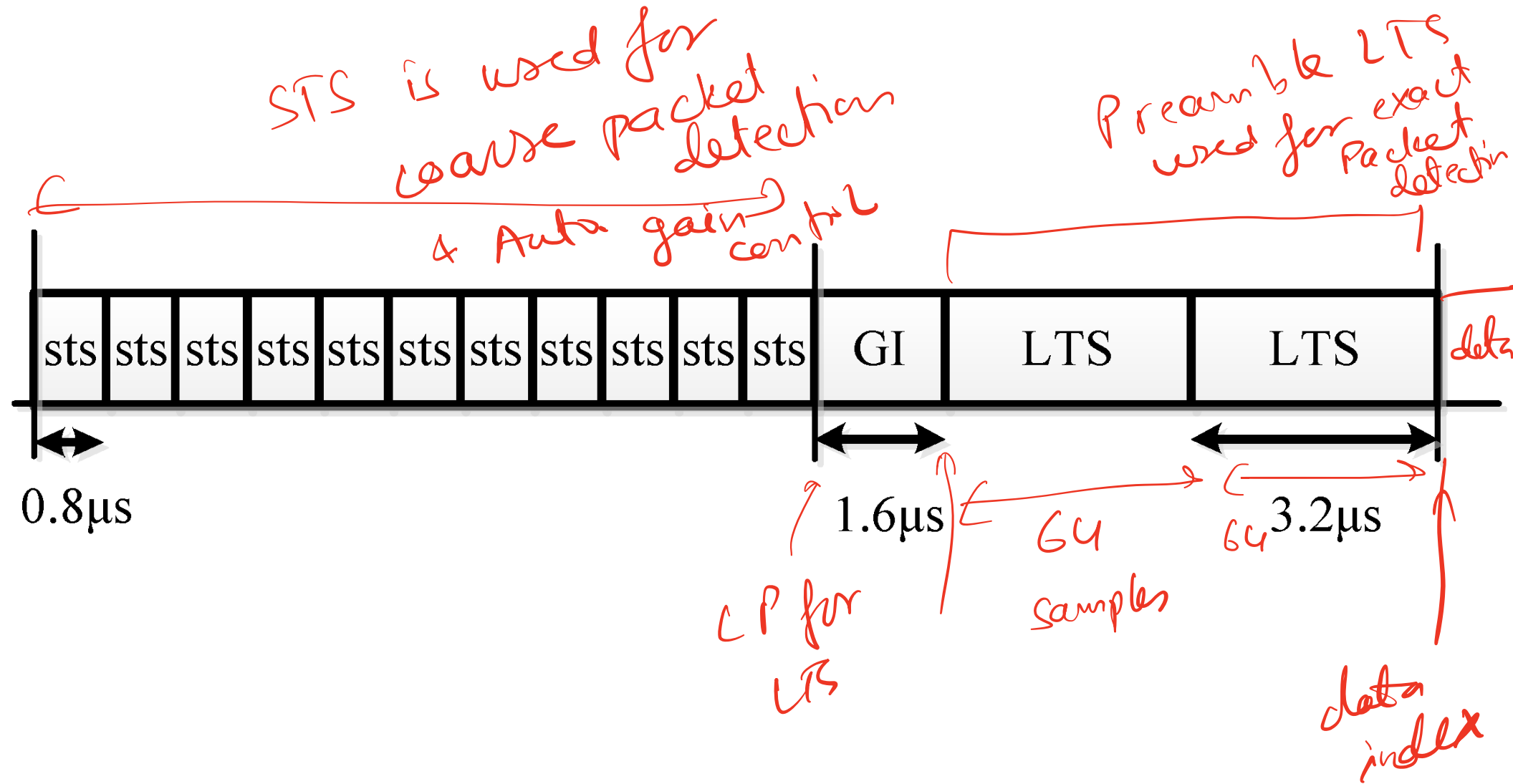
$$f_d = \frac{f_c v}{c}$$



Design

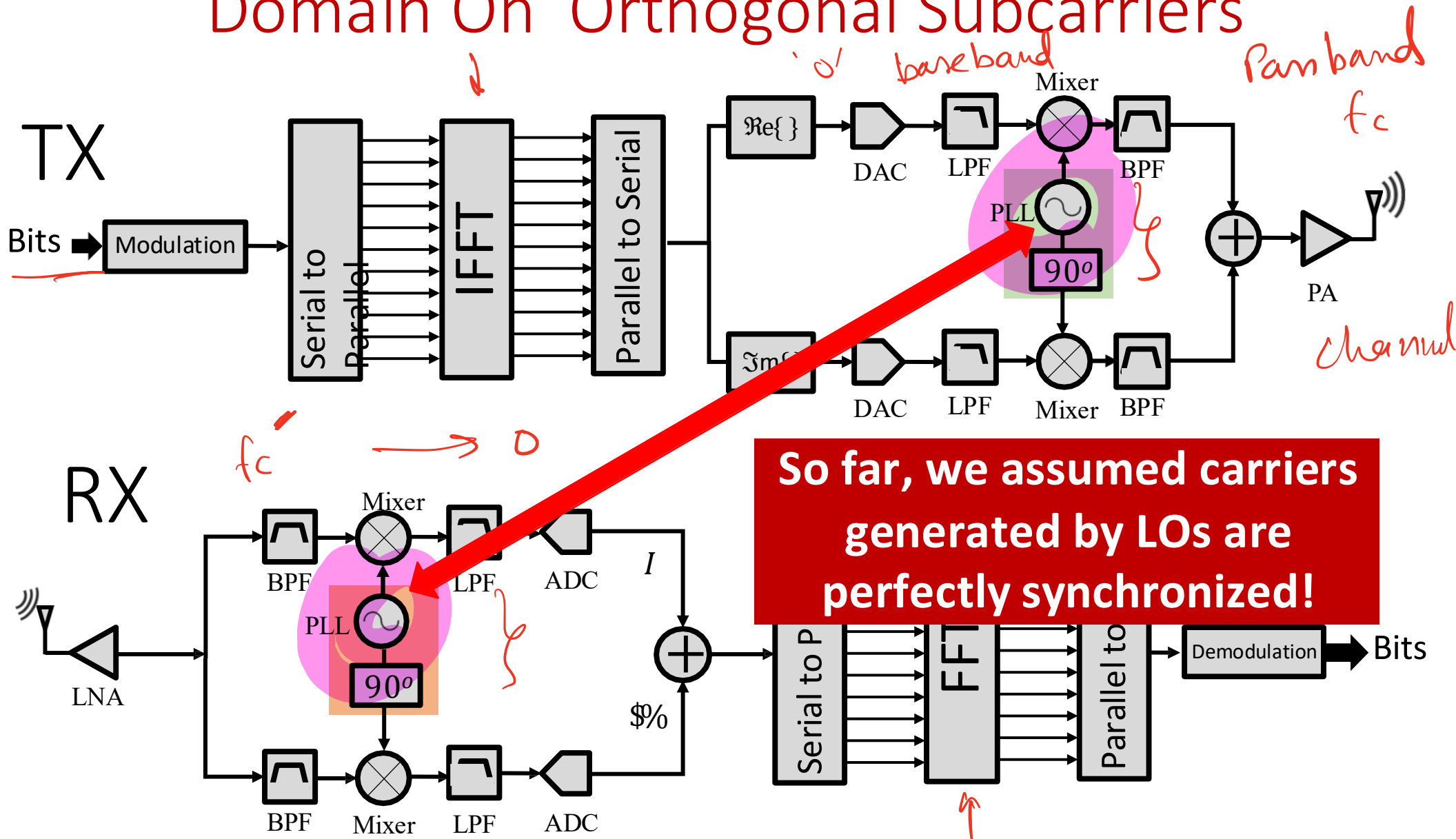


WiFi Example: STS and LTS



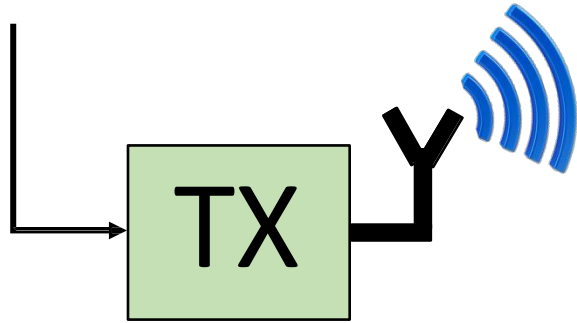
OFDM: Orthogonal Frequency Division

Multiplexing **Transmit Symbols in Frequency Domain On Orthogonal Subcarriers**

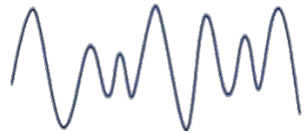


Carrier Frequency Offset

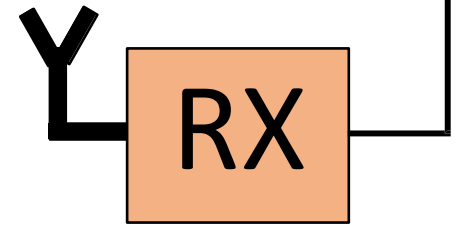
101101011



if $f_c' = f_c$
(perfect sync.)



101101011

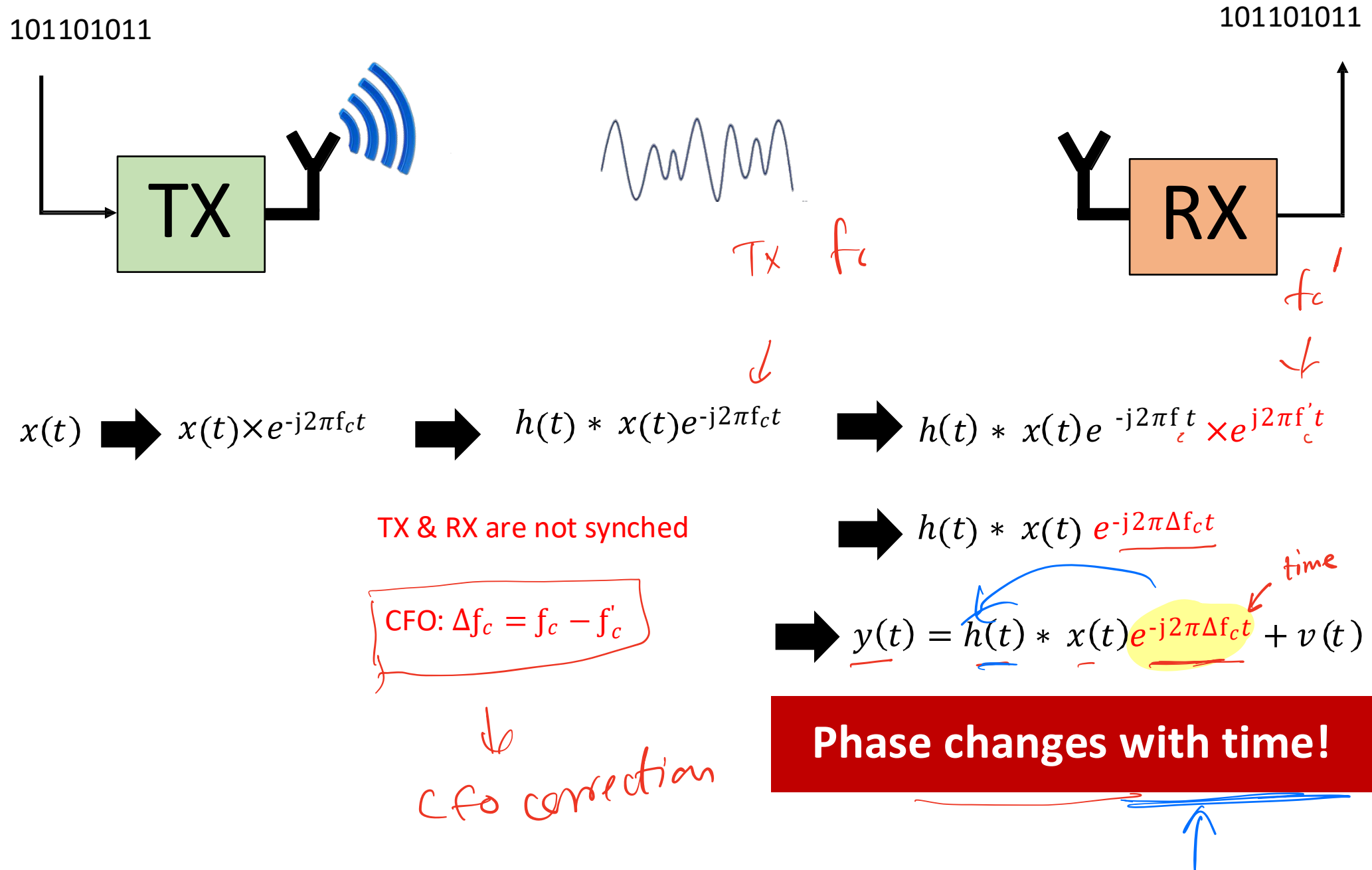


Recall

$$\begin{aligned}
 \underline{x(t)} &\rightarrow \underline{x(t) \times e^{-j2\pi f_c t}} \xrightarrow{\text{upconvert}} h(t) * x(t) e^{-j2\pi f_c t} \xrightarrow{\text{apply channel}} h(t) * x(t) e^{-j2\pi f_c t} \times \cancel{e^{j2\pi f_c t}} \\
 &\xrightarrow{\text{downconvert}} \underline{h(t) * x(t)} \xrightarrow{\text{noise}} y(t) = h(t) * x(t) + v(t)
 \end{aligned}$$

Assumes TX & RX perfectly synched

Carrier Frequency Offset

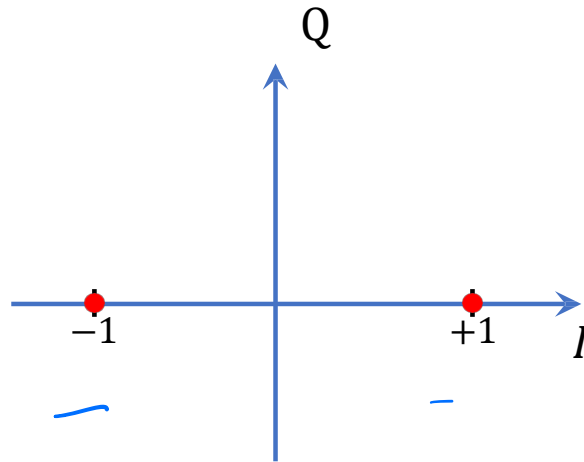


Carrier Frequency Offset

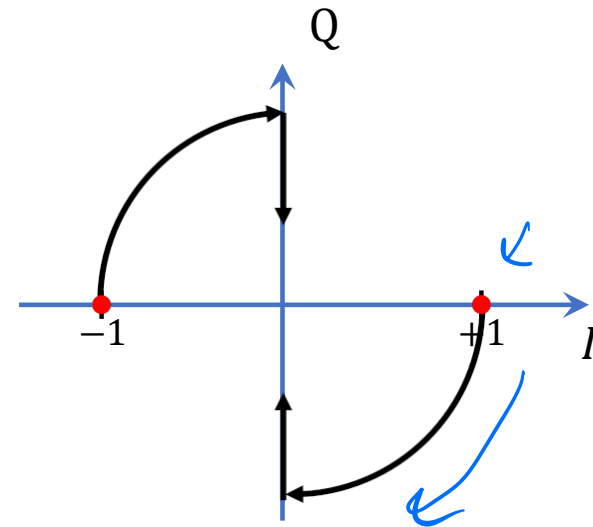
Consider BPSK Modulation.

$0 \rightarrow -1$

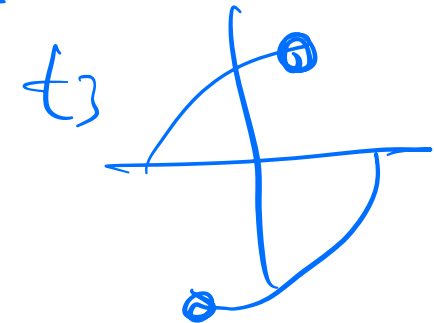
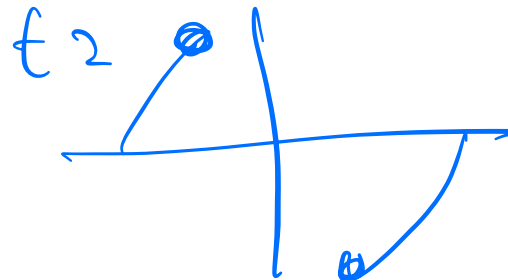
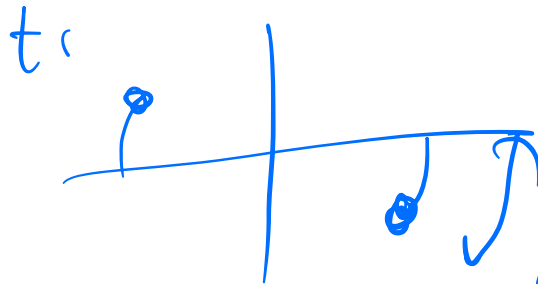
$1 \rightarrow +1$



$x(t)$



$h x(t - r) \underline{e^{-j2\pi\Delta f_c t}} + v(t)$



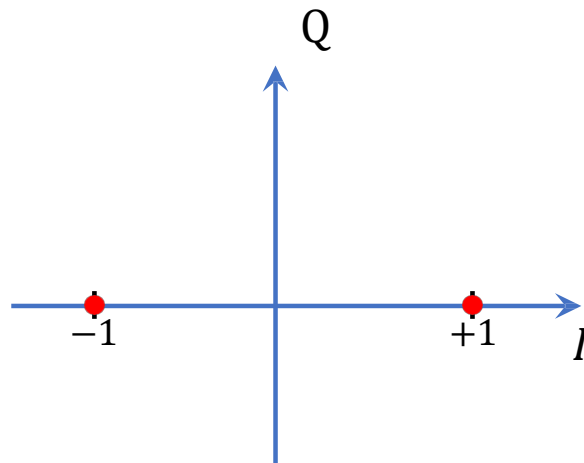
$2\pi\Delta f_c t$

Carrier Frequency Offset

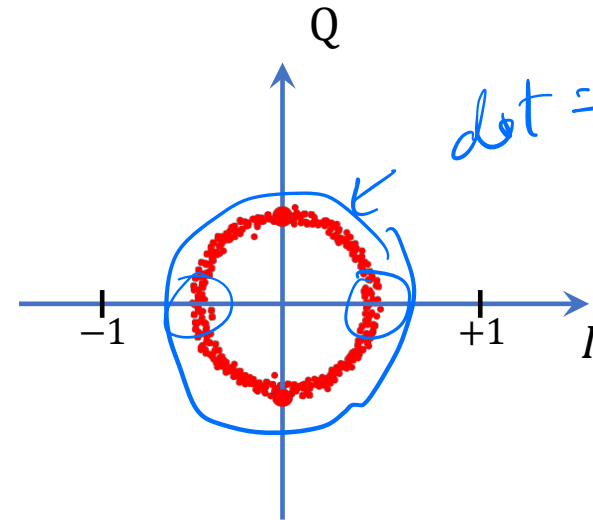
Consider BPSK Modulation.

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$x(t)$



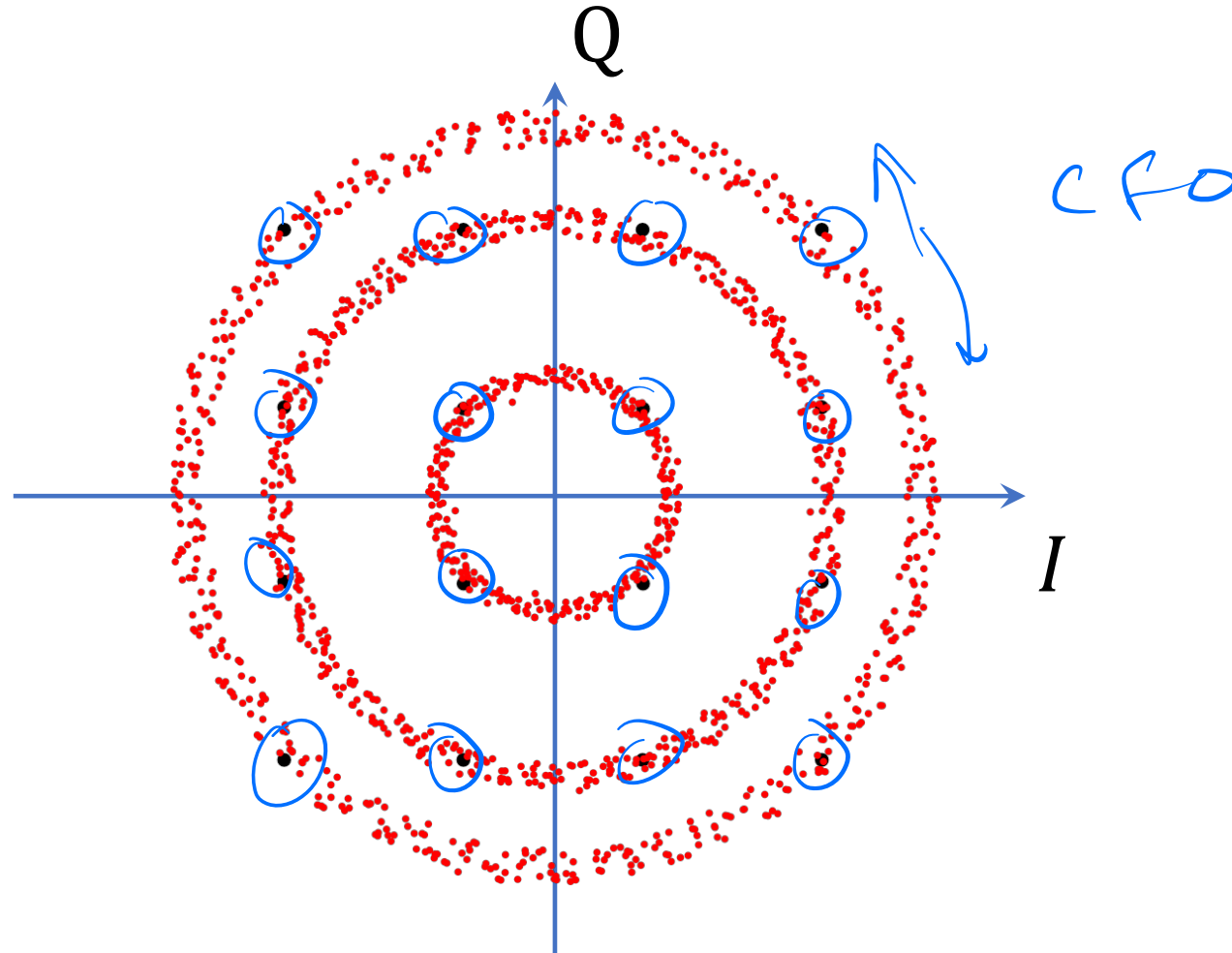
$$h x(t - r) e^{-j2\pi\Delta f_c t} + v(t)$$

Impossible to Decode!

$\Rightarrow$ BER will be high

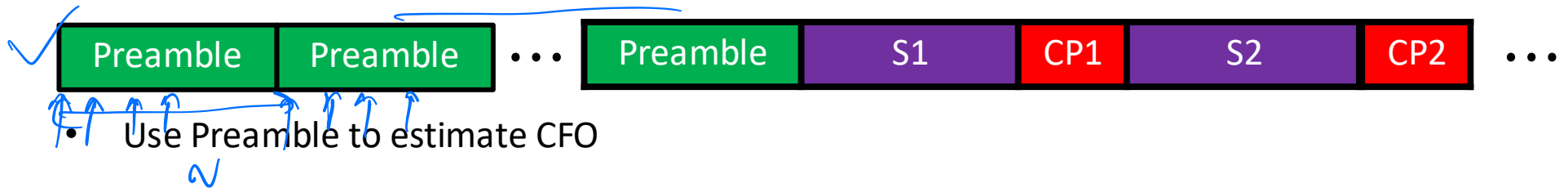
Carrier Frequency Offset

Consider 16 QAM Modulation



Need to estimate and correct CFO to decode!

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1(t) = x(t)e^{-j2\pi\Delta f_c t}$$

Sample at $t = nT_s$

Symbol has N samples $\rightarrow T = NT_s$

$$y_2(t) = x(t)e^{-j2\pi\Delta f_c (t+T)}$$

$$\frac{y_2(t)}{y_1(t)}$$

Handwritten derivation for CFO estimation:

$$t_1, t_2$$

$$2\pi\Delta f_c t_1 \quad \& \quad 2\pi\Delta f_c t_2$$

is known

$$\Delta f_c = \frac{2\pi\Delta f_c t_2 - 2\pi\Delta f_c t_1}{2\pi(t_2 - t_1)}$$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1[n] = x[n]e^{-j2\pi\Delta f_c nT_s}$$

$$y_2[n] = x[n]e^{-j2\pi\Delta f_c (nT_s + NT_s)}$$

Sample at $t = nT_s$

Symbol has N samples $\rightarrow T = NT_s$

subcarriers
FFT size
samples in one OFDM symbol

= # samples in Preamble

same for all n

- Compute:

$$A = \sum_{t=1}^N \underbrace{y_1[n] y_2^*[n]}_{\text{LTS1 LTS2}} = \sum_{t=1}^N \underbrace{x[n] x^*[n]}_{\text{same for all } n} e^{j2\pi\Delta f_c NT_s}$$

$$A = e^{j2\pi\Delta f_c NT_s} \sum_{t=1}^N |x[n]|^2$$

can recover Δf_c

$t=1$

constant

$$\Delta f_c = \frac{\angle A}{2\pi NT_s}$$

OFDM CFO Estimation & Correction



- Use Preamble to estimate CFO

$$y_1[n] = x[n]e^{-j2\pi\Delta f_c nT_s}$$

$$y_2[n] = x[n]e^{-j2\pi\Delta f_c (nT_s + NT_s)}$$

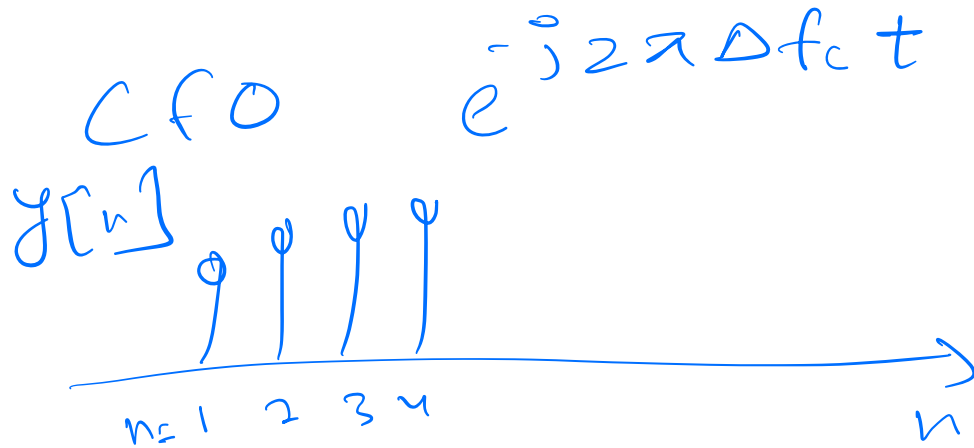
- Compute:

$$A = \sum_{t=1}^N y_1[n]y_2^*[n] \rightarrow \Delta f_c = \frac{\angle A}{2\pi nT_s}$$

- Correct CFO:

$$y[n] \times e^{j2\pi\Delta f_c nT_s}$$

entire y
entire $x \dots$ end of packet



$$t = nT_s$$

$$T_s = \frac{1}{f_c}$$

$$e^{-j2\pi\Delta f_c nT_s}$$

CFO is too high

$$\Delta f_c T_s > 1$$

$$\Delta f_c > \frac{1}{T_s}$$

$\Rightarrow$ can't detect it reliably

$$\text{Phase} = 2\pi\Delta f_c nT_s \bmod 2\pi$$

$$2\pi + \delta \bmod 2\pi = \delta$$

at t

$$2\pi\Delta f_c t$$

$t + NT_s$

$$2\pi\Delta f_c (t + NT_s)$$

$$2\pi\Delta f_c (t + NT_s) - 2\pi\Delta f_c t$$

Phase estimate = $2\pi\Delta f_c NT_s \bmod 2\pi$

$$\Delta f_c \times NT_s < 1 \Rightarrow \text{detected CFO reliably}$$

OFDM Channel Estimation



- Use Preamble to estimate the channel

$$\underline{y(t)} = \underline{h(t)} * \underline{x(t)} \leftrightarrow Y(f) = \underline{H(f)} X(f)$$

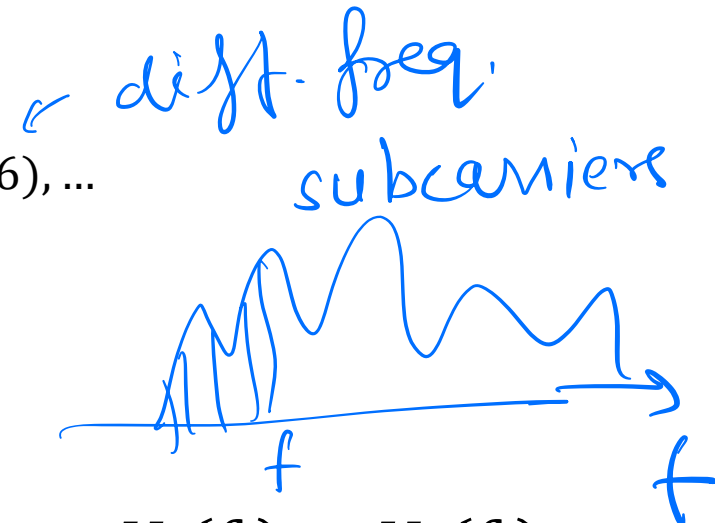
↑

- Send $X(f)$: $\underline{-1}, \underline{+1}, \underline{-1}, \underline{-1}, \underline{-1}, \underline{+1}, \dots$

- Receive: $\underline{-H(1)}, \underline{H(2)}, \underline{-H(3)}, \underline{-H(4)}, \underline{-H(5)}, \underline{H(6)}, \dots$

- Estimate:

$$\tilde{H}(f) = \frac{Y(f)}{X(f)}$$



- Use two preambles to average noise:

$$\tilde{H}(f) = \frac{Y_1(f) + Y_2(f)}{2 X(f)}$$

$Y_1(f) = 2TS1$
 $Y_2(f) = 2TS2$



↓

Y_1

↓

Y_2

↓

$$Y_d = H X_d$$

$$H = \frac{1}{2} \left(\frac{Y_1}{X} + \frac{Y_2}{X} \right)$$

$$X_d = \frac{Y_d}{H}$$

Phase Tracking

So Far: Estimated and Corrected For Coarse Value of CFO

- Residual CFO:

$$y(t) = h(t) * x(t) e^{j2\pi\Delta f_c t} + v(t)$$

$$\Delta f_c = df_c + \delta f_c$$

Coarse CFO

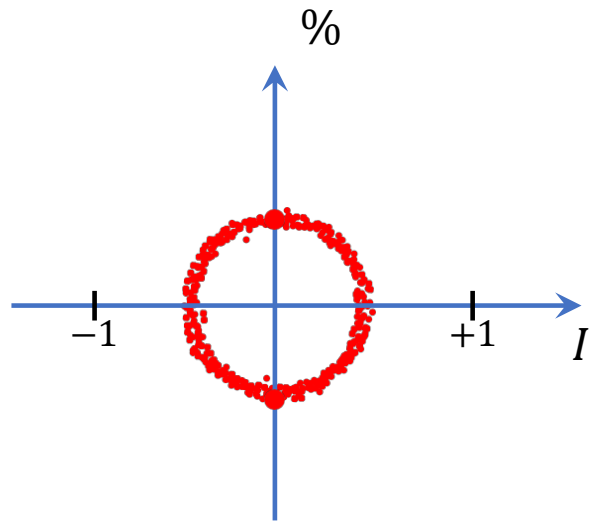
Residual CFO

We estimated and corrected for coarse CFO!

Even small residual can accumulate over time
to create large phase: $e^{-j2\pi\delta f_c t}$

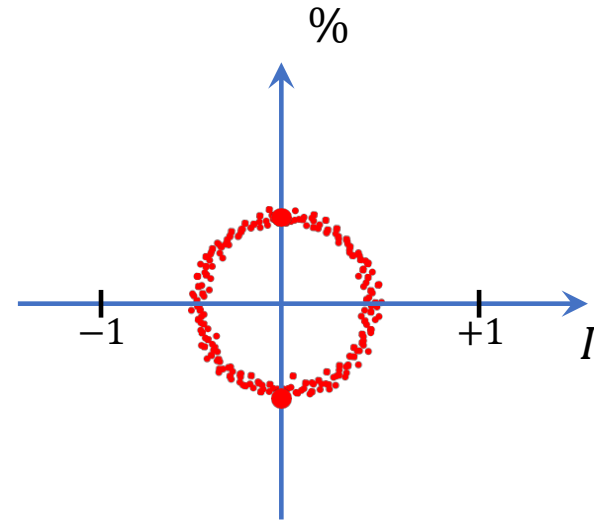
Need to track the phase

Phase Tracking



$$h x(t-r) e^{j2\pi\Delta f_c t} + v(t)$$

Without Course CFO Correction



$$h x(t-r) e^{j2\pi\delta f_c t} + v(t)$$

Without Residual CFO Correction

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

When we sample the signal there is a residual sampling offset: $n\delta T_s$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

When we sample the signal there is a residual sampling offset: $n\delta T_s$

$$y[n] = x[n + n\delta T_s] e^{-j2\pi\delta f_c n T_s}$$

$$Y[f] = X[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi\delta f_c n T_s}$$

Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)

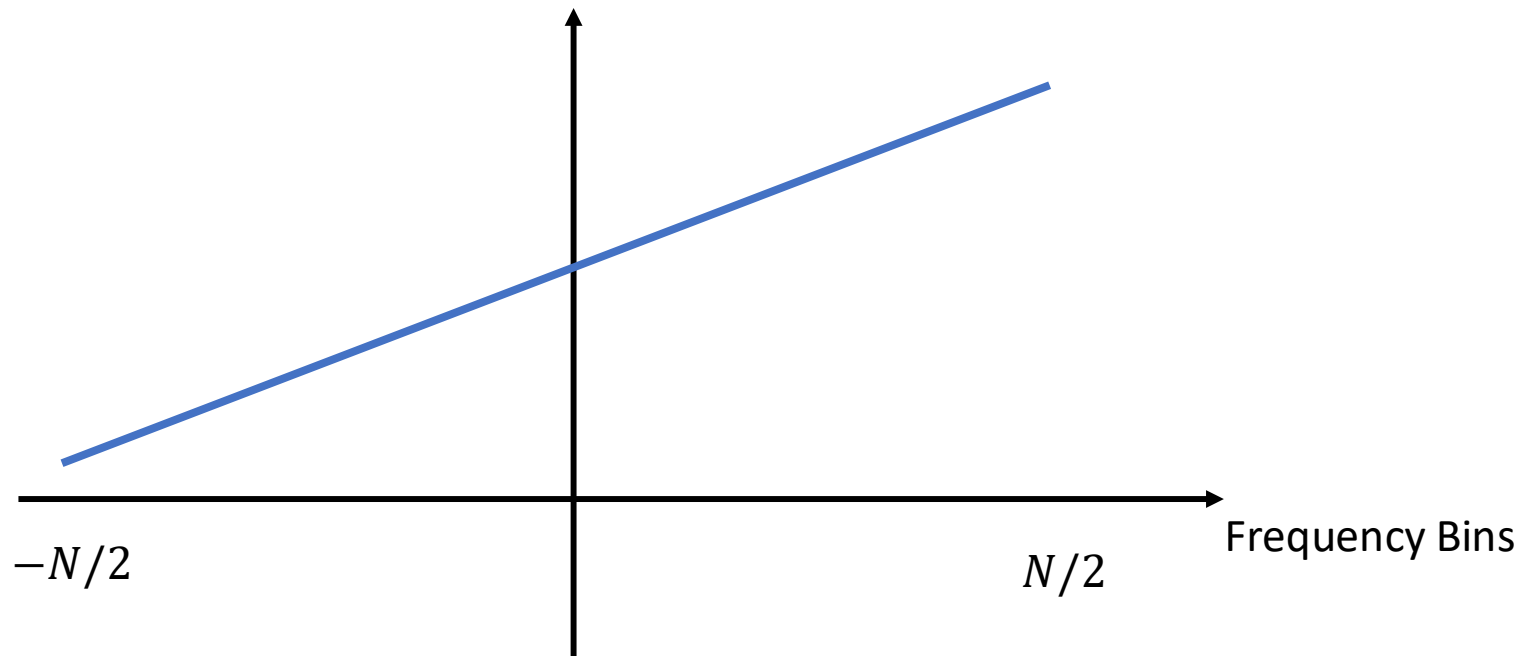
$$Y_1[f] = X_1[f] e^{j \frac{2\pi f n \delta T_s}{N} - 2\pi \delta f_c n T_s}$$

$$Y_2[f] = X_2[f] e^{j \frac{2\pi f (n+N+CP) \delta T_s}{N} - 2\pi \delta f_c (n+N+CP) T_s}$$

Phase accumulation: $\Delta\phi = 2\pi f \frac{(N+CP)\delta T_s}{N} - 2\pi \delta f_c (N+CP) T_s$

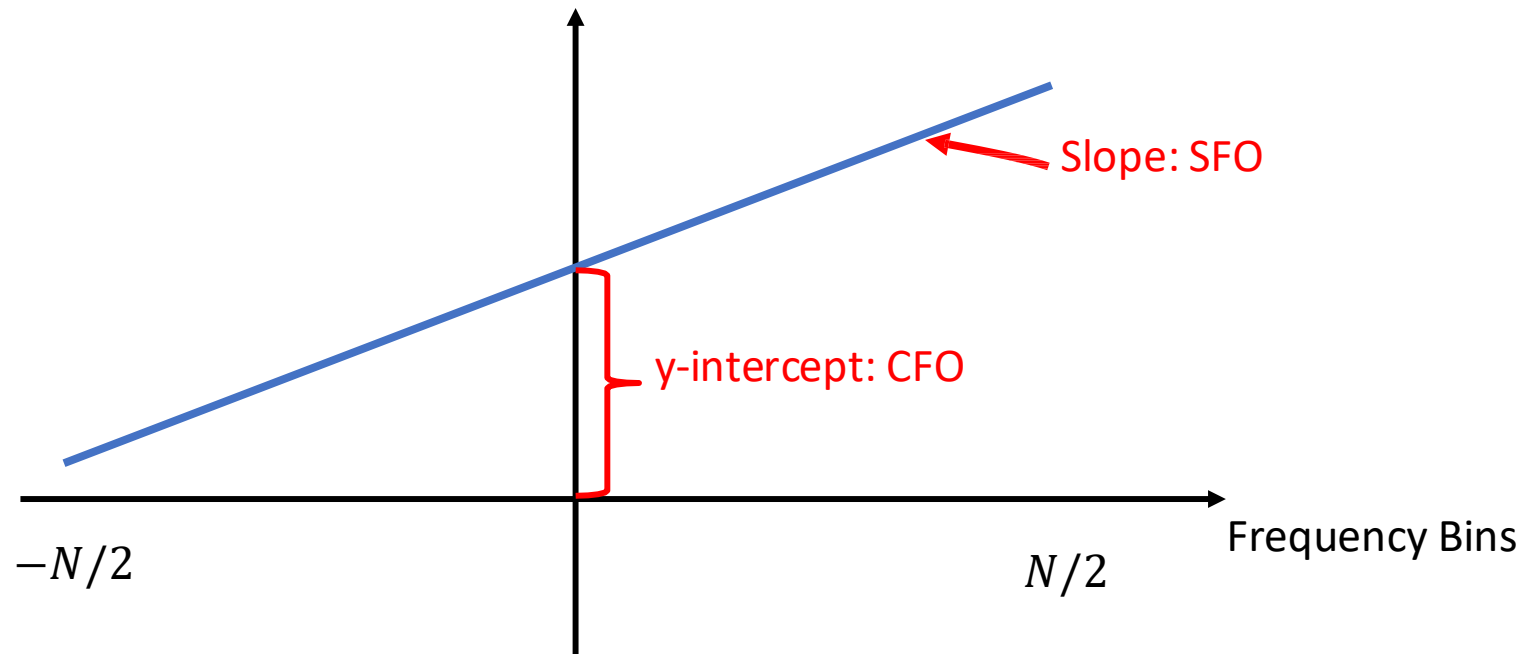
Phase Tracking

- Residual CFO (Carrier Frequency Offset)
- Residual SFO (Sampling Frequency Offset)
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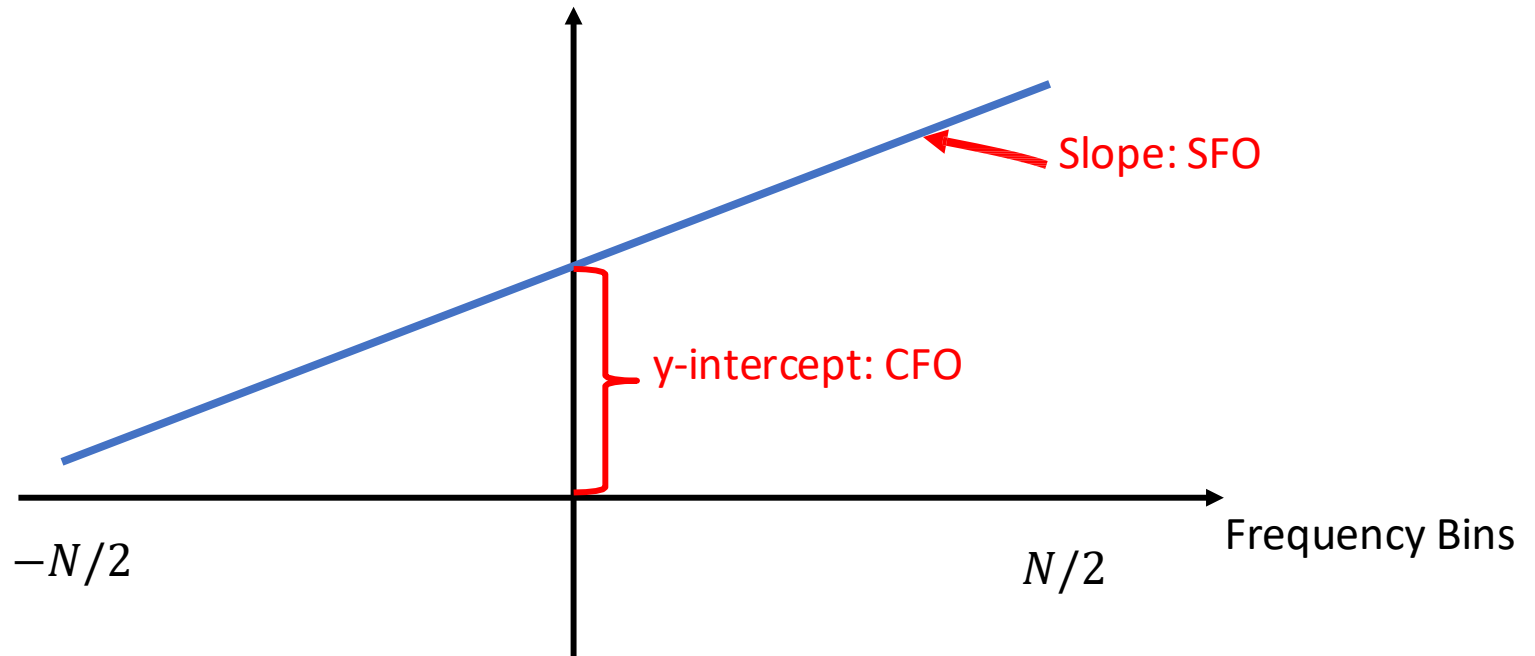
Phase Tracking

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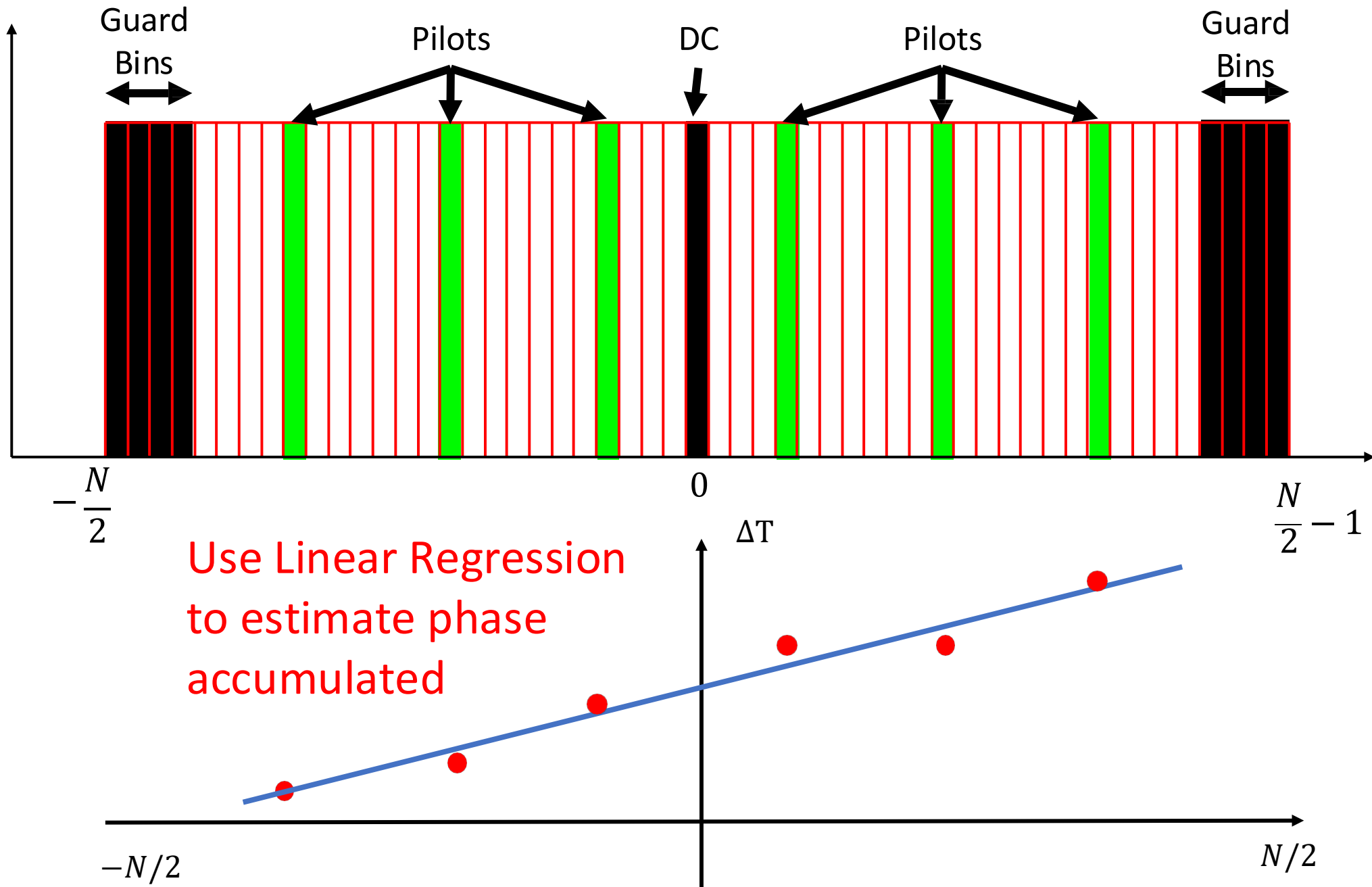
OFDM Phase Tracking

- Phase accumulation: $\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$



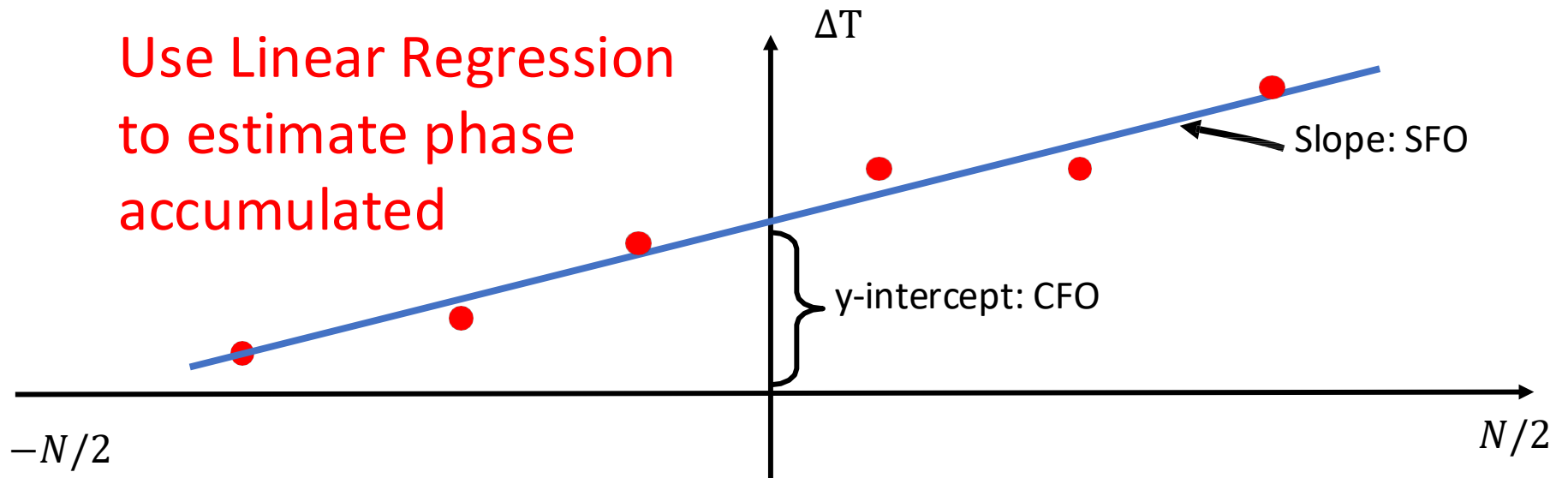
- Sufficient to estimate slope & y-intercept to know the phase accumulated for all subcarriers.
- Use only few subcarriers as pilots & send known bits in them.

OFDM Phase Tracking



OFDM Phase Tracking

Use Linear Regression
to estimate phase
accumulated

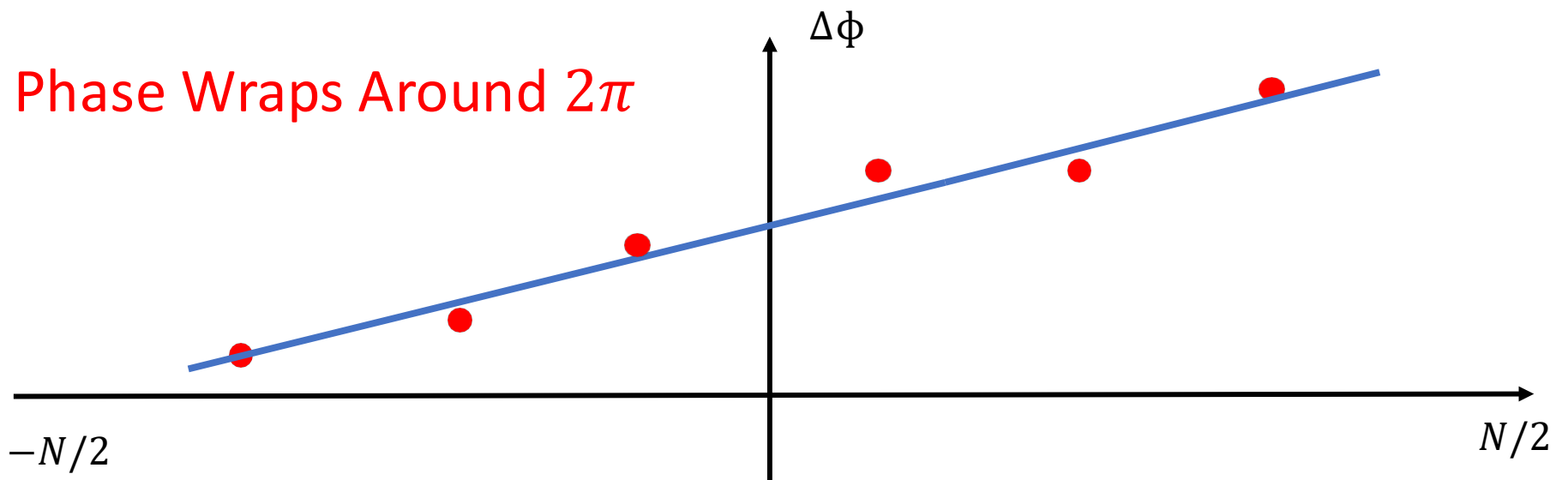


$$\text{Slope} = 2\pi \frac{(N + CP)\delta T_s}{N}$$

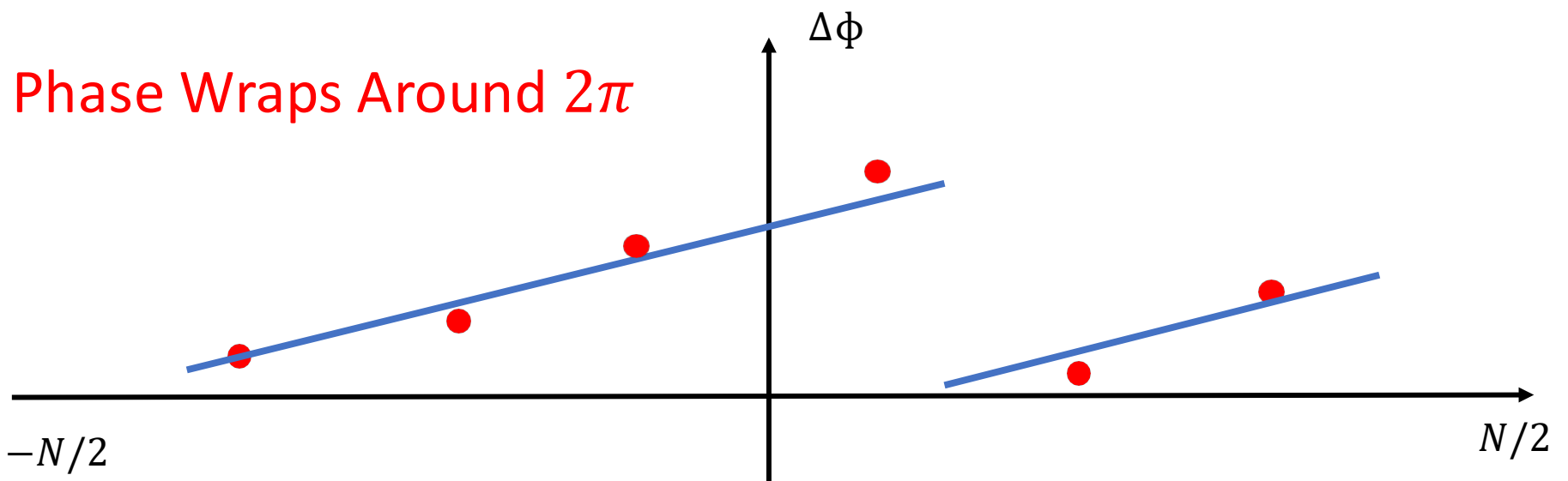
$$Y - \text{intercept} = -2\pi\delta f_c(N + CP)T_s$$

$$\Delta\phi = 2\pi f \frac{(N + CP)\delta T_s}{N} - 2\pi\delta f_c(N + CP)T_s$$

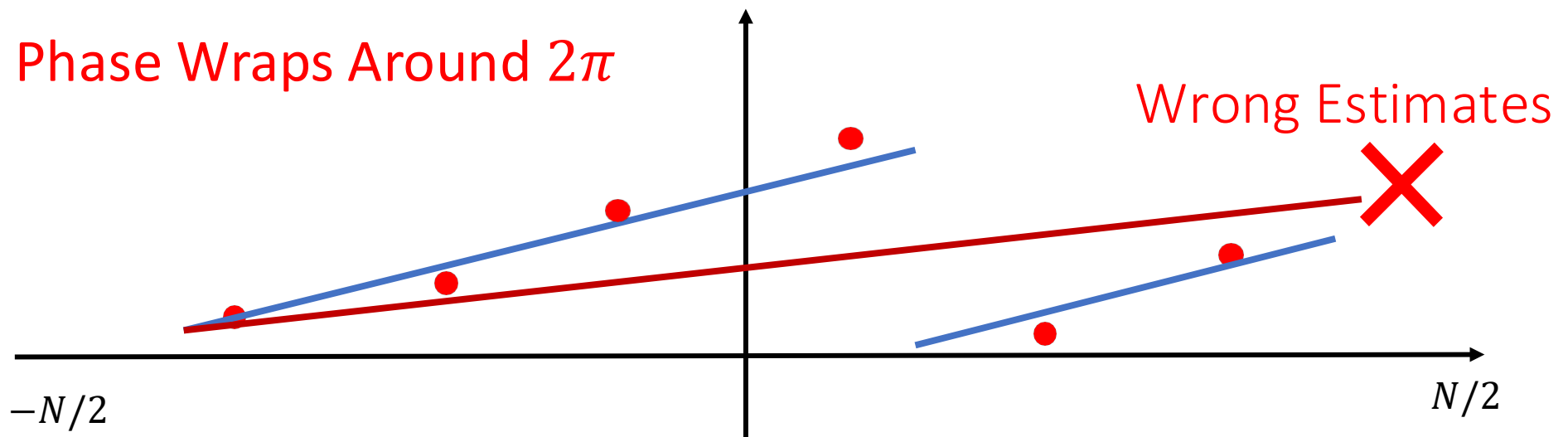
Be Careful!



Be Careful!



Be Careful!

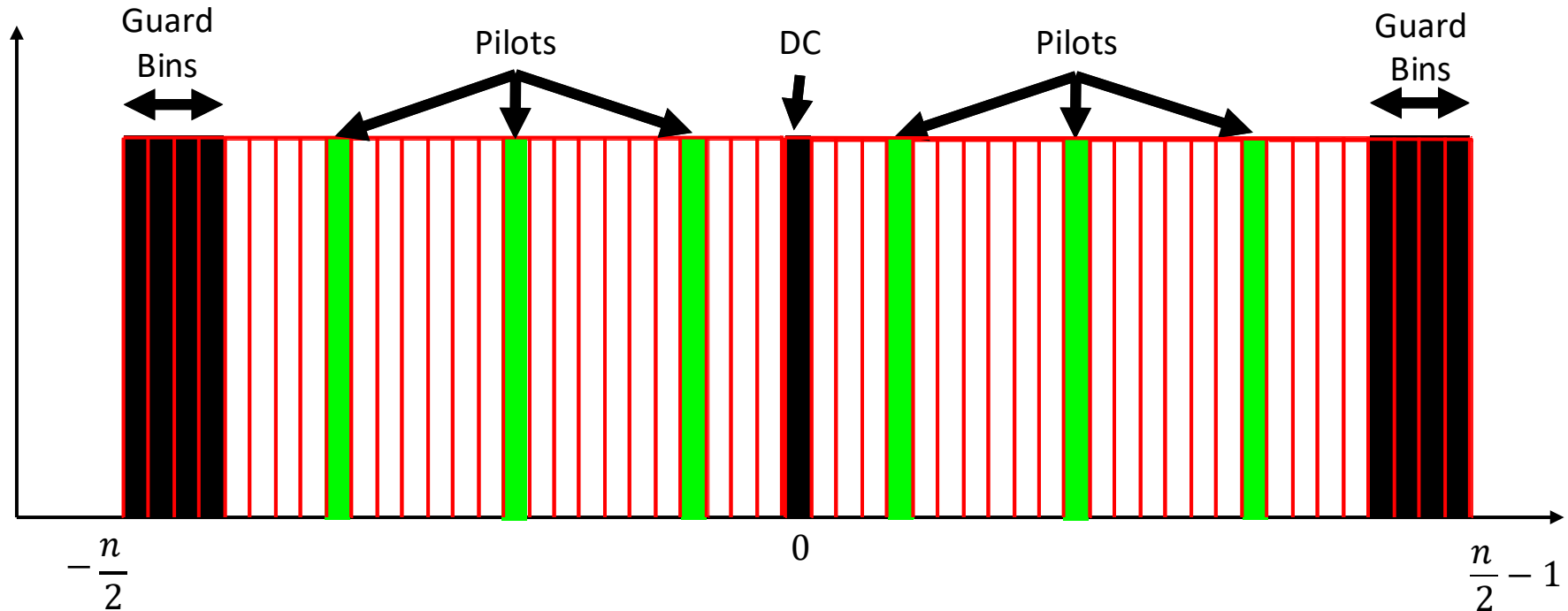


Need to unwrap phase mod 2π first! Use `unwrap()` function in MATLAB

When estimating slope and y-intercept, equation assumes subcarriers index f is between $-n/2$ and $n/2 - 1$

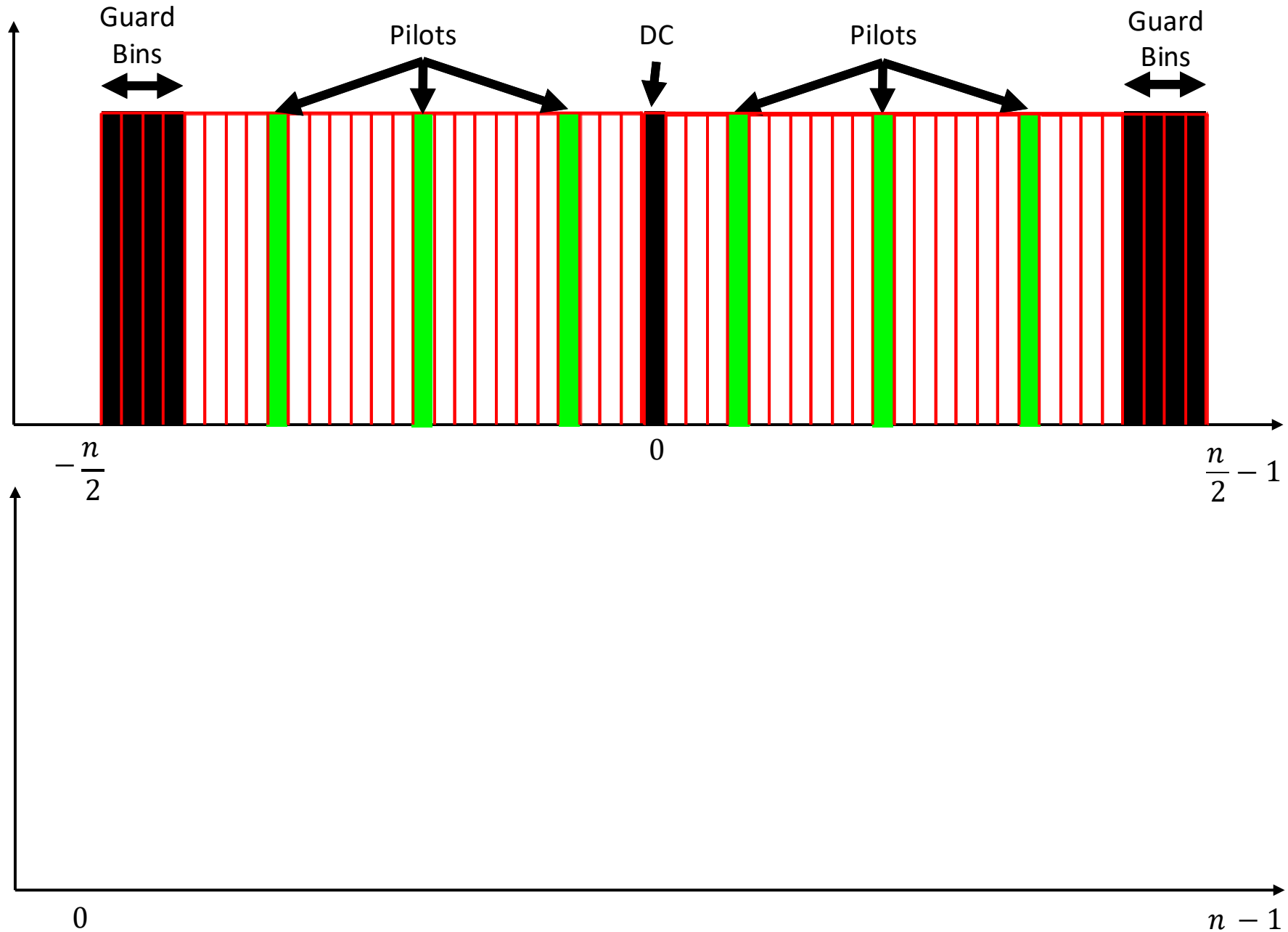
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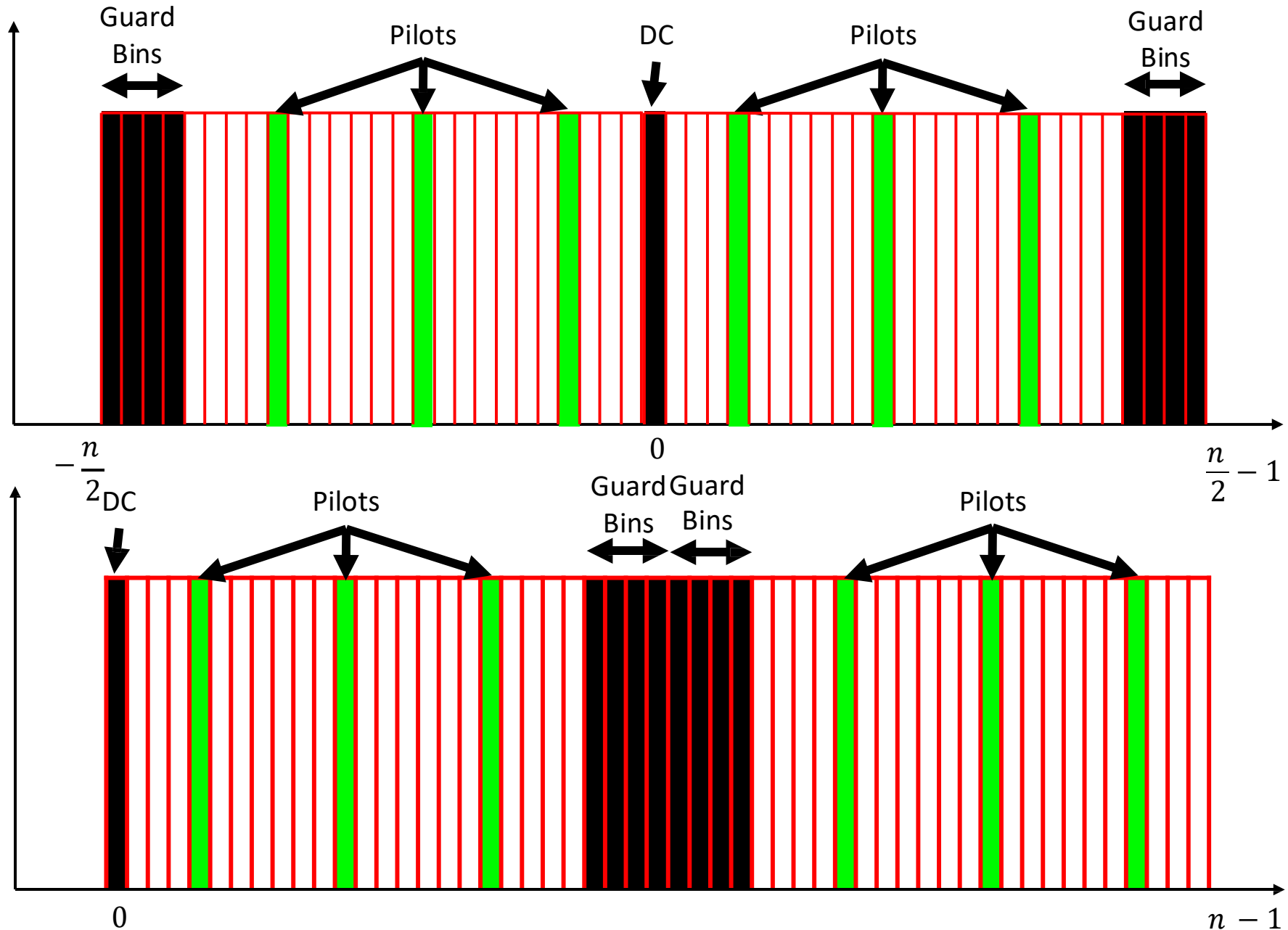
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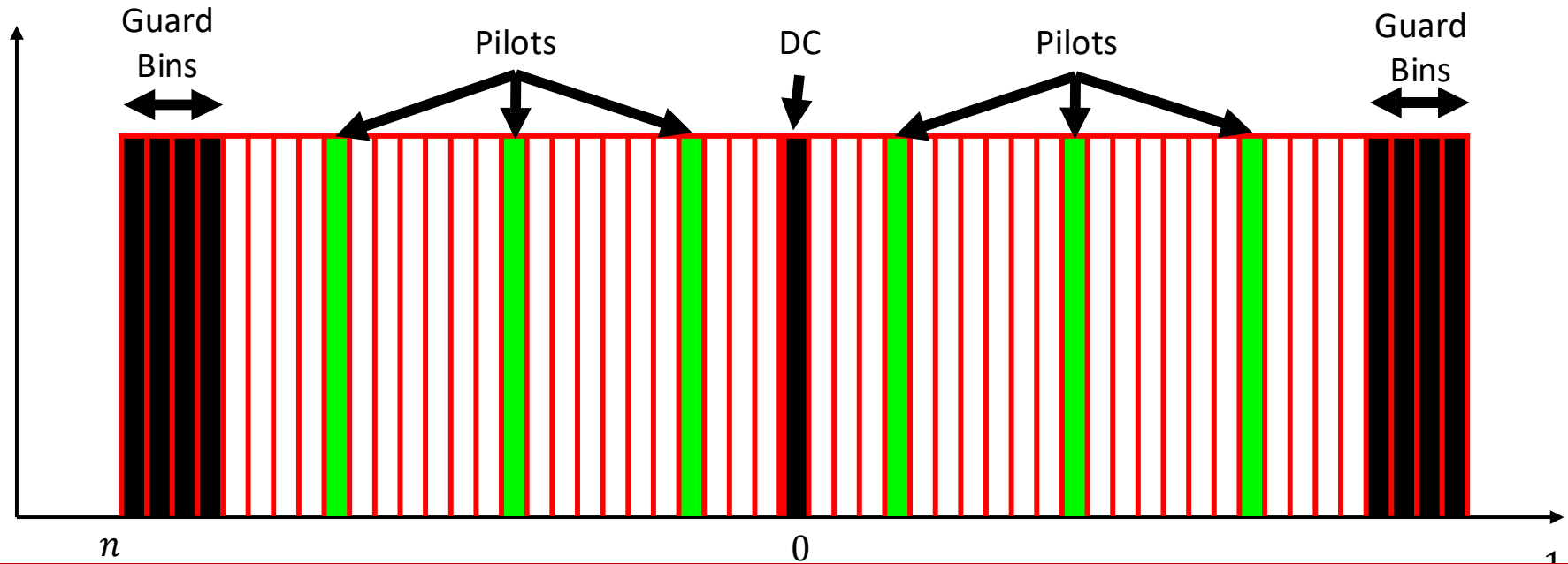
Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index + is between $-\frac{n}{2}$ and $\frac{n}{2} - 1$



Be Careful!

When estimating slope and y-intercept, equation assumes subcarriers index + is between $-n/2$ and $n/2 - 1$



Need to `fftshift()` function in MATLAB + provided functions to convert indexing.

